VLM-Powered CDS for Handwritten Prescriptions: Research Summary

Background and problem

Most outpatient prescriptions in Bangladesh are handwritten and dispensed with no automated safety check. Clinical decision support (CDS), which flags interactions and dosing errors in high-income settings, depends on structured digital data that paper prescriptions lack. Traditional OCR fails on cursive handwriting, abbreviations and mixed Bangla-English notation, so the CDS layer never receives usable input.

Research questions

1. How accurately can Vision Language Models (VLMs) extract drug name, dose, route and frequency from handwritten Bangladeshi prescriptions?
2. How do extraction errors propagate into downstream CDS alerts (missed and false hazards)?
3. What share of real outpatient prescriptions contain preventable hazards a VLM-CDS pipeline could flag?

Methodology

The study runs a four-stage pipeline on de-identified prescriptions from a Bangladeshi private hospital.

1. **Image capture:** anonymized prescription scans, annotated by clinicians as ground truth.
2. **VLM extraction:** VLMs read images and output structured medication data (drug, strength, dose, frequency, duration).
3. **Normalization:** brand names mapped to generic ingredients using a Bangladesh drug database.
4. **CDS screening:** rule-based checks for drug-drug interactions, dose limits and therapeutic duplication.

Evaluation: field-level extraction accuracy (precision, recall, F1) against annotations; hazard-detection sensitivity and false-alert rate against clinician review.

Open: final VLMs compared, sample size and annotation protocol.

Contributions and significance

- **Empirical:** first local estimate of preventable prescribing hazards in handwritten outpatient prescriptions at this setting.
- **Technical:** a benchmark of VLMs on low-resource, mixed-script medical handwriting.
- **Safety:** quantifies how extraction errors affect CDS reliability, informing human-in-the-loop design.
- **Policy:** evidence for medication-safety investment where digital prescribing is still rare.

Application and future work

The pipeline complements Duha Rx, Duha Health's chamber management software for Bangladeshi doctors. Duha Rx enables real-time CDS on digital prescriptions; the VLM pipeline extends checks to paper prescriptions still in circulation, at pharmacies or on intake. Next steps: integrate validated CDS rules into Duha Rx, add Bangla-language support, and release a public screening tool with clear accuracy disclaimers.